EXPERIMENT 10: GIT BRANCHING AND MERGE CONFLICT RESOLUTION

Aim

To demonstrate Git branching and merge conflict resolution by creating a feature branch, making conflicting changes to the same file on two branches, resolving the conflict, and verifying the final repository status.

Procedure

1. Switch to the main branch, pull the latest changes from origin, and verify that the working tree is clean.

2. Create a feature branch named feature-conflict-demo.

3. Modify README.md in the feature branch by adding the Branch Conflict Demonstration section.

4. Stage, commit, and push the feature-branch changes to GitHub.

5. Switch to main and modify the same README section with different text.

6. Stage, commit, and push the main-branch change to GitHub.

7. Switch back to feature-conflict-demo and merge main to trigger a conflict in README.md.

8. Resolve the conflict by replacing the conflicting text with a single final statement and removing the conflict markers.

9. Stage README.md, commit the conflict resolution, and push the resolved branch.

10. Run git status and git log --oneline --graph --all -5 to verify the clean working tree and review the commit history.

Output

The following screenshots show the feature branch creation, changes committed on both branches, the conflict-resolution commit, and final Git verification.

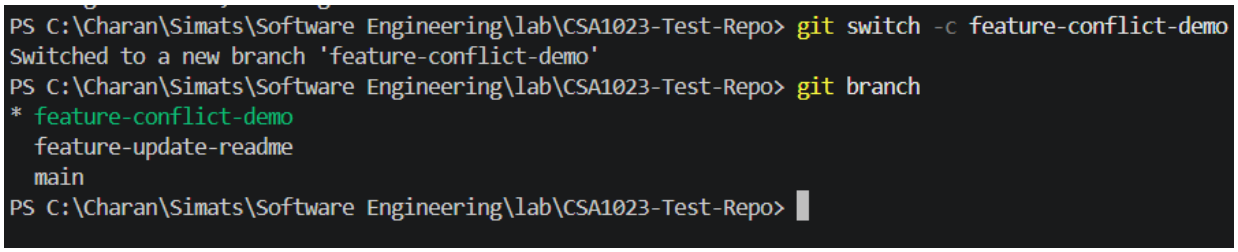


Figure 1: Created the feature-conflict-demo branch from main.

```
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git add README.md
>> git commit -m "Update README in feature branch"
>> git push -u origin feature-conflict-demo
[feature-conflict-demo 4c27f5b] Update README in feature branch
 1 file changed, 4 insertions(+), 1 deletion(-)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 391 bytes | 391.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-conflict-demo' on GitHub by visiting:
remote:      https://github.com/charankumarReddyB/CSA1023-Test-Repo/pull/new/feature-conflict-demo
remote:
To https://github.com/charankumarReddyB/CSA1023-Test-Repo.git
 * [new branch]      feature-conflict-demo -> feature-conflict-demo
branch 'feature-conflict-demo' set up to track 'origin/feature-conflict-demo'.
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo>
```

Figure 2: Committed and pushed the feature-branch README update.

```
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git add README.md
>> git commit -m "Update README in main branch"
>> git push origin main
[main 602ef44] Update README in main branch
 1 file changed, 4 insertions(+), 1 deletion(-)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 381 bytes | 381.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/charankumarReddyB/CSA1023-Test-Repo.git
   e1931b8..602ef44  main -> main
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo>
```

Figure 3: Committed and pushed a different README update on main.

```
Your branch is up to date with 'origin/feature-conflict-demo'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git add README.md
>> git commit -m "Resolve README merge conflict"
>> git push origin feature-conflict-demo
[feature-conflict-demo d944a23] Resolve README merge conflict
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 399 bytes | 399.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/charankumarReddyB/CSA1023-Test-Repo.git
   4c27f5b..d944a23  feature-conflict-demo -> feature-conflict-demo
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo>
```

Figure 4: Resolved the README merge conflict, committed the resolution, and pushed the branch.

```
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git status
On branch feature-conflict-demo
Your branch is up to date with 'origin/feature-conflict-demo'.

nothing to commit, working tree clean
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git log --oneline --graph --all -5
*   d944a23 (HEAD -> feature-conflict-demo, origin/feature-conflict-demo) Resolve README merge confli
ct
|\
| * 602ef44 (origin/main, origin/HEAD, main) Update README in main branch
* | 4c27f5b Update README in feature branch
|/
*   e1931b8 Merge pull request #1 from charankumarReddyB/feature-update-readme
|\
| * 85178d1 (origin/feature-update-readme, feature-update-readme) Update README with contribution wor
kflow
|/
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo>
```

Figure 5: Verified a clean working tree and reviewed the Git commit graph.

Result

Thus, Git branching and merge conflict resolution were successfully demonstrated. Conflicting edits to README.md were identified and resolved, the resolution was committed and pushed, and the repository was verified to have a clean working tree.